# Experiment 4

## Control Structure using Conditional Statements

## Aim / Objective:

The objective of this experiment is to help students understand and apply control structures using conditional statements in programming. Through this experiment, students will learn how conditional statements such as if, if-else, if-else-if ladder, and switch are used to control the logical flow of a program based on specific conditions. They will gain the ability to choose and implement the appropriate conditional structure depending on the problem requirements. Students will also develop the skills to trace and debug the flow of execution in programs that involve decisionmaking. This experiment will encourage students to write clean, efficient, and readable code by applying best practices in conditional logic..

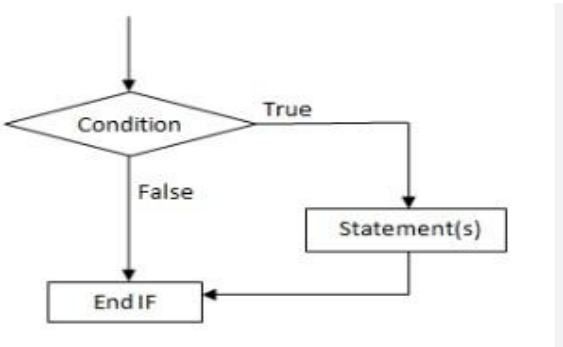
## Problem Statement:

1. **Write a program to find positive,negative and zero using if-else if (multi-way decision).**
2. **Write a program to find whether a year is a leap year or not using if-else.**

## Theory / Concept:

In programming, control structures are essential elements that define the flow of execution in a program. Conditional statements are a type of control structure that allows a program to make decisions based on certain conditions. These decisions determine which set of instructions should be executed under specific circumstances

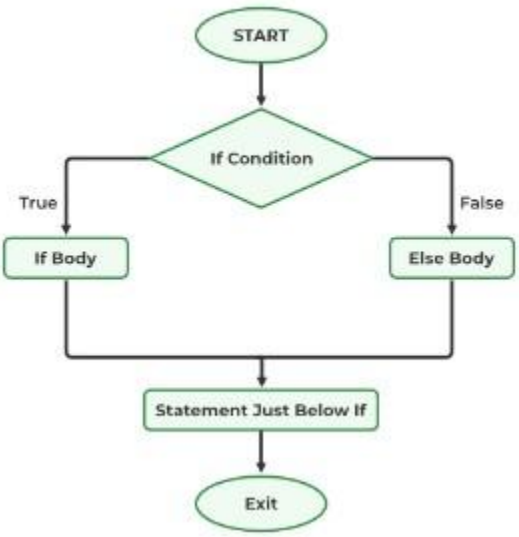
The most commonly used conditional structures are:
*1. if* ***Statement***
The if statement checks a condition and executes a block of code **only if** the condition is true. **Flowchart:**



*2. **if-else Statement***

The if-else statement provides an **alternate path**: one for when the condition is true, and another for when it is false.

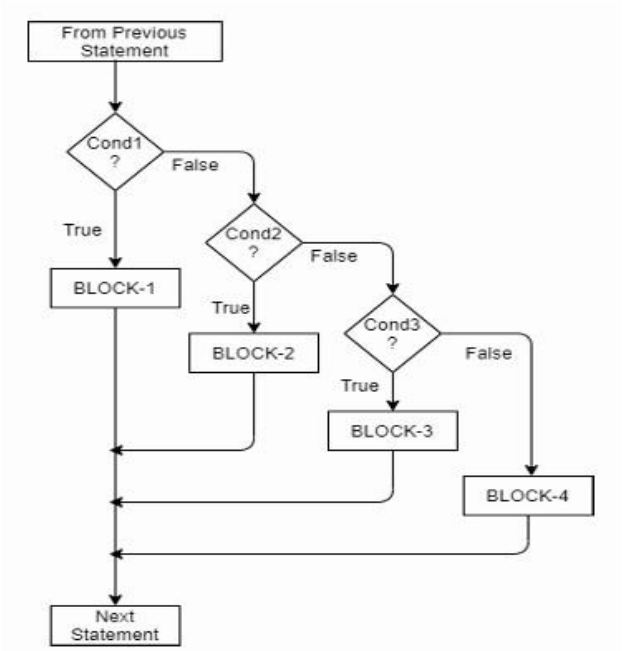
**Flowchart:**




*3. **if-else-if Ladder***

Used when there are **multiple conditions** to evaluate in sequence.

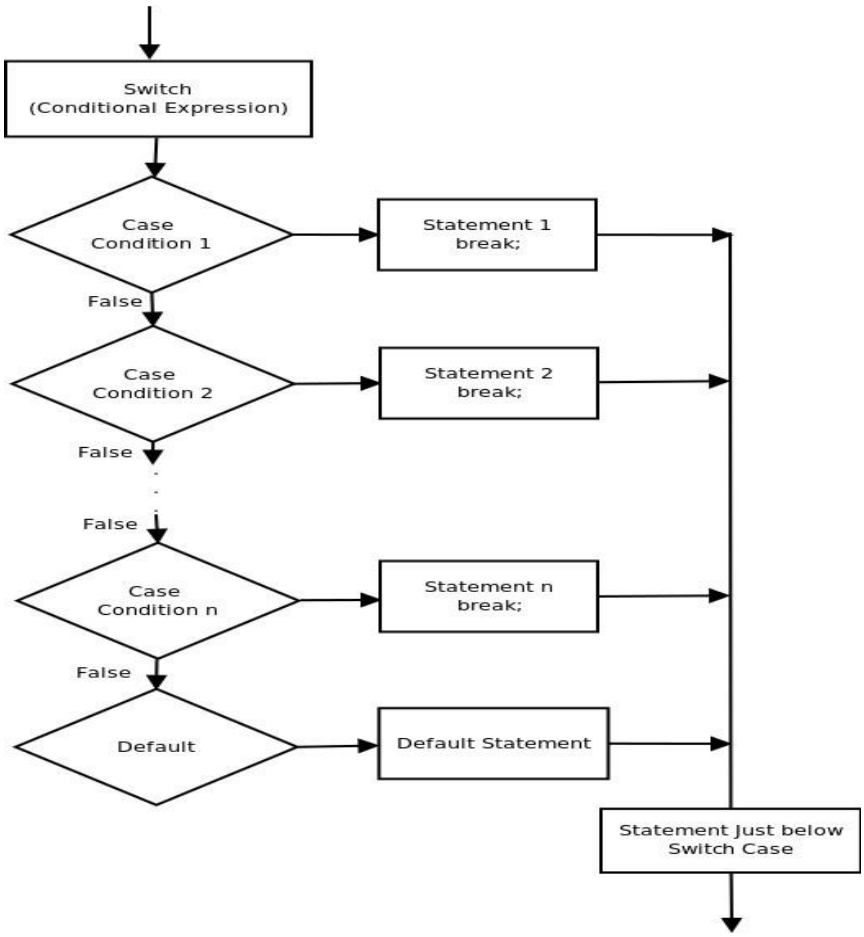
**Flowchart:**



*4.* ***switch Statement***

Used when a variable is compared against **multiple constant values**. Each value is called a *case*.

**Flowchart:**




These conditional structures often use **relational operators** (==, !=, >, <, >=, <=) and **logical operators** (&&, ||, !) to form expressions. These expressions determine which part of the code should be executed.

Mastering conditional control structures helps students write programs that respond to different inputs and conditions intelligently. This foundational skill is essential for developing logicdriven applications such as menu systems, calculators, login validation, and decision-based systems.

## Program code:

| |
|---|
| **1. A program to find positive,negative and zero using if-else if** |

**Algorithm:**

Step 1: Start
Step 2: Declare a variable to store the number (e.g., num)
Step 3: Read the value of num from the user Step
4:
If num > 0, then
Display "The number is positive" Step
5:
Else if num < 0, then
Display "The number is negative" Step
6:
Else
Display "The number is zero"
Step 7: End

## Flowchart:

**Code:**

```
#include <stdio.h>
int main()
{
   int num;
   printf("Enter a number: ");
scanf("%d", &num);
   if (num > 0)
   {
      printf("The number is positive.\n");
   }
   else if (num < 0)
   {
      printf("The number is negative.\n");
   }
else
   {
      printf("The number is zero.\n");
   }
   return 0;
}
```

**Output:**

main.c | Output

```
Enter a year: 2024
2024 is a leap year.


=== Code Exited With Errors ===
```

**2. Write a program to find whether a year is a leap year or not using if-else.**

**Algorithm:**

1. **Start**
2. **Input** a year from the user and store it in variable `year`
3. **Check** the following conditions:
   - If `year` is divisible by 4
     → Then check if it is divisible by 100
     → If yes, check if it is divisible by 400
     → If yes, it's a leap year
     → Else, it's **not** a leap year
     → If not divisible by 100, it's a leap year
   - Else (not divisible by 4), it is **not** a leap year
4. **Display** the result
5. **End**

**Flowchart:**

**Code:**

```
#include <stdio.h> void
main()
{
   int year;
printf("Enter a year: ");
scanf("%d", &year);
   if (year % 4 == 0)
   {
      if (year % 100 == 0)
      {
         if (year % 400 == 0)
         {
            printf("%d is a leap year.\n", year);
         } else
         {
            printf("%d is not a leap year.\n", year);
         }
}
else
      {
         printf("%d is a leap year.\n", year);
      }
}
else
   {
      printf("%d is not a leap year.\n", year);
}
}
```

# Output:

main.c | Output

```
Enter a year: 2024
2024 is a leap year.


=== Code Exited With Errors ===
```

**Practice Programs:**

1. **Write a program** to input a quiz score out of 10 and print a message:
   a. 10: "Excellent!"
   b. 7–9: "Good job"
   c. 4–6: "Try harder"
   d. Below 4: "Poor"

2. **Write a program** to simulate a **traffic light system** using `switch` where the input is a color (`Red`, `Yellow`, or `Green`) and the output is the appropriate driving instruction.

## Compilation and Execution Steps:

1. Open Dev C++.
2. Click File → New → Source File.
3. Type or paste your C code.
4. Click File → Save As and save the file with a .c extension.
5. Click Execute → Compile (or press **F9**).
6. If there are no errors, click Execute → Run (or press **Ctrl + F10**).
7. View output in the console window.

**Learning Outcomes:** Students will able to LO3.1
Define loop control structures.
LO3.2 Illustrate the use of switch cases.
LO3.3 Illustrate the different types of ternary and conditional operators.

**Course Outcome:** After completion of course the students will be able to various Control Structures in C.

## Conclusion:

In this experiment, we learned how to use conditional statements (if, if-else, if-else-if) to control program flow based on different conditions. We successfully implemented programs to check if a number is positive, negative, or zero, and to determine if a year is a leap year. This helped us understand decision-making in C programming..

**For Faculty Use only**

| Correction Parameters | Formative Assessment [40%] | Timely completion of Practical [ 40%] | Attendance / Learning Attitude [20%] | Total |
|---|---|---|---|---|
| Marks Obtained | | | | |